

The Program in Action

When you run *collatz.py*, the output will look like this:

```
Collatz Sequence, or, the 3n + 1 Problem
By Al Sweigart al@inventwithpython.com
```

```
The Collatz sequence is a sequence of numbers produced from a starting
number n, following three rules:
```

```
--snip--
```

```
Enter a starting number (greater than 0) or QUIT:
```

```
> 26
```

```
26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1
```

```
Collatz Sequence, or, the 3n + 1 Problem
By Al Sweigart al@inventwithpython.com
```

```
--snip--
```

```
Enter a starting number (greater than 0) or QUIT:
```

```
> 27
```

```
27, 82, 41, 124, 62, 31, 94, 47, 142, 71, 214, 107, 322, 161, 484, 242, 121,
364, 182, 91, 274, 137, 412, 206, 103, 310, 155, 466, 233, 700, 350, 175, 526,
263, 790, 395, 1186, 593, 1780, 890, 445, 1336, 668, 334, 167, 502, 251, 754,
377, 1132, 566, 283, 850, 425, 1276, 638, 319, 958, 479, 1438, 719, 2158,
1079, 3238, 1619, 4858, 2429, 7288, 3644, 1822, 911, 2734, 1367, 4102, 2051,
6154, 3077, 9232, 4616, 2308, 1154, 577, 1732, 866, 433, 1300, 650, 325, 976,
488, 244, 122, 61, 184, 92, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5,
16, 8, 4, 2, 1
```

How It Works

The % mod operator can help you determine if a number is even or odd. Remember that this operator is a sort of “remainder” operator. While 23 divided by 7 is 3-remainder-2, 23 mod 7 is simply 2. Even numbers divided by 2 have no remainder, while odd numbers divided by 2 have a remainder of 1. When *n* is even, the condition in `if n % 2 == 0`: on line 33 evaluates to True. When *n* is odd, it evaluates to False.

```
1. """Collatz Sequence, by Al Sweigart al@inventwithpython.com
2. Generates numbers for the Collatz sequence, given a starting number.
3. More info at: https://en.wikipedia.org/wiki/Collatz_conjecture
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: tiny, beginner, math"""
6.
7. import sys, time
8.
9. print('Collatz Sequence, or, the 3n + 1 Problem
10. By Al Sweigart al@inventwithpython.com
11.
12. The Collatz sequence is a sequence of numbers produced from a starting
13. number n, following three rules:
14.
15. 1) If n is even, the next number n is n / 2.
16. 2) If n is odd, the next number n is n * 3 + 1.
```